

Math 421: Homework 11

Solutions

”Due”: Dec 12, 2025

Problem 3: (Ross 29.5) - Solution given in back of textbook.

Problem 8: (Ross 29.11) Show $\sin x \leq x$ for all $x \geq 0$.

Proof. Let $f(x) = x - \sin x$. Then $f'(x) = 1 - \cos x$, since $\cos x \leq 1$, we have $f'(x) \geq 0$ for all x . Thus, $f(x)$ is an increasing function on the interval $[0, \infty)$ and therefore achieves its minimum value at $f(0) = 0 - \sin 0 = 0$, thus, for all $x \in (0, \infty)$, we have $f(x) = x - \sin x \geq 0$, so that $\sin x \leq x$. At 0, we have $0 = x = \sin x = 0$, so $\sin x \leq x$ for all $x \in [0, \infty)$. $\square$

Problem 5: (Ross 29.17) - Solution given in back of textbook.

Problem 7: (Ross 32.6) - Let f be a bounded function on $[a, b]$. Suppose there exist sequences (U_n) and (L_n) of upper and lower Darboux sums for f such that $\lim(U_n - L_n) = 0$. Show f is integrable and $\int_a^b f = \lim U_n = \lim L_n$.

Proof. Let (U_n) and (L_n) be sequences of upper and lower Darboux sum, then $U_n = U(f, P_n)$ and $L_n = L(f, Q_n)$ for some partitions P_n and Q_n of $[a, b]$. Since $\lim(U_n - L_n) = 0$, we have for every $\epsilon > 0$ there exists N such that for all $n > N$ we have $|U_n - L_n| < \epsilon$. Consider the partition $P_n \cup Q_n$, we have

$$L(f, Q_n) \leq L(f, P_n \cup Q_n) \leq U(f, P_n \cup Q_n) \leq U(f, P_n)$$

so that

$$U(f, P_n \cup Q_n) - L(f, P_n \cup Q_n) \leq U(f, P_n) - L(f, Q_n) = U_n - L_n < \epsilon.$$

Therefore, for every $\epsilon > 0$ there exists a partition P such that

$$U(f, P) - L(f, P) < \epsilon$$

and f is integrable on $[a, b]$ by the Cauchy criterion of integrability.

Since f is integrable, we know $L(f) = U(f) = \int_a^b f$ and therefore

$$L_n \leq L(f) = \int_a^b f = U(f) \leq U_n$$

and

$$0 \leq \int_a^b f - L_n \leq U_n - L_n \text{ and } L_n - U_n \leq \int_a^b f - U_n \leq 0$$

Therefore, $\lim L_n = \int_a^b f = \lim U_n$. $\square$

Problem 1: (Ross 33.1) - Solution given in back of textbook.

Problem 1: (Ross 33.4) - Hint given in back of textbook. Try to think of your own example!